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MILITARY SPECIFICATION

OMELET WITH SMOKED SAUSAGE AND POTATOES, ARTIFICIALLY COLORED, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers omelet with smoked sausage and potatoes, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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listed in the issue of [illegible] of the documents which are DoD adopted are those

MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit; Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-44340 - Can, Tray Pack

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies may be obtained from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59)

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

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U.S. Standards for Condition of Food Containers

(Copies may be obtained from the Chairman, Condition of Food Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0608, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Copies should be obtained from the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Copies should be obtained from the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

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its territories, or storage facilities located within the United States

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Copies should be obtained from the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Sausage links, smoked, cooked. Cooked, smoked sausage links shall be commercially cured in accordance with USDA Meat and Poultry Inspection Regulations. The sausage links shall contain not more than 90 percent pork and not less than 10 percent beef based on raw meat weight. Added water shall not exceed 4 times the protein content plus 10 percent as determined by USDA methodology. The cooked, smoked sausage links shall not contain more than 2.2 percent corn syrup or corn syrup liquid equivalent. The cooked, smoked sausage links shall have a fat content of not greater than 30 percent and a salt content of not greater than 2.7 percent. Each sausage link shall be approximately 2 inches ($\pm$ 1/8 inch) in length and the count per pound shall be 46 to 52. The cooked, smoked sausage links shall have a uniform brown surface color and shall possess a heavy smoke flavor. Packaged sausage links shall be maintained at an internal temperature of 28° to 40°F for not more than 15 days prior to tray pack can filling. If destined for freezing, the packaged sausage links shall be frozen to an internal temperature of 0°F or below within 72 hours after packaging and shall be held for not more than 30 days prior to tray pack can filling and sealing.

3.2.2 Water. Water used for formulation, rehydration, blanching, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

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3.2.3 Citric acid. Citric acid shall comply with the Food Chemicals Codex.

3.2.4 Color, annatto, dry. Annatto color shall be a reddish-orange water soluble powder of the extractive of Annatto bixa Orellana. The powder shall contain not more than 5.0 percent moisture and 100 percent of the powder shall pass through a U.S. Standard No. 10 sieve. The total color (as percent Norbixin) shall be 15 percent.

3.2.5 Whole eggs, liquid or frozen. Whole eggs may by liquid or frozen and shall have been processed and labeled in accordance with the Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59). The whole eggs shall be egg whites and egg yolks in their natural proportions as broken directly from the shell eggs as evidenced by a USDA Egg Products Inspection Certificate. For liquid whole eggs, the certificate shall state the date and time of pasteurization. Liquid whole eggs shall be held at a temperature of 40°F or lower and shall be held for not more than 72 hours from the time of pasteurization until the start of formulation of the product in which the liquid whole eggs are used. Frozen whole eggs shall be held at 10°F or lower and used within 120 days. The whole eggs shall be free from off-odors, off-flavors, or foreign odors and flavors such as metallic, musty, sour, fruity, or sulfide-like, and shall be free from foreign materials.

3.2.6 Potatoes, dehydrofrozen diced. Dehydrofrozen potatoes shall be firm, possess a good color and flavor, and be of a white flesh variety suitable for canning. The maximum specific gravity for the potatoes shall be 1.075 with a reducing sugar content of not more than 2.0 percent on a dry weight basis. The potatoes shall be mechanically diced so that the dimensions of the dices will be approximately 3/8 by 3/8 by 3/8 inch when rehydrated.

3.2.7 Potatoes, fresh. Potatoes shall be fresh, clean, sound, and of a white flesh variety suitable for canning. The maximum specific gravity for the potatoes shall be 1.075 with a reducing sugar content of not more than 2.0 percent on a dry weight basis.

3.2.8 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.9 Starch, waxy maize, modified, pregelatinized, instant. The starch shall be a pure, off-white, odorless, slightly coarse ground powder. It shall be modified, pregelatinized, cross-linked, and stabilized. The starch shall have a maximum moisture content of 9.0 percent and a pH of 4.0 to 6.5. The starch shall be resistant to breakdown under high temperature heating conditions, shall possess good low temperature stability, and shall not set to a gel. The starch shall display good clarity and a smooth, short textured, heavy body. The starch shall be bland with no objectionable off-notes. The starch shall comply with MIL-STD-900.

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3.2.10 Oil, vegetable. Vegetable oil shall possess a clean, bland flavor and shall have a minimum stability of 15 hours (A.O.M), a free fatty acid value not to exceed 0.05 percent, a moisture/volatile matter content not to exceed 0.06 percent, and shall pass a cold test of 5.5 hours (minimum).

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Preparation of potatoes.

3.3.1.1 Preparation of fresh potatoes. The fresh potatoes shall be thoroughly cleaned to remove dark and extraneous material, peeled, trimmed, and thoroughly washed. The washed potatoes shall be mechanically diced with dicer settings at approximately 3/8 by 3/8 by 3/8 inch. The potato dices shall be blanched sufficiently to prevent discoloration and to remove excess air. The blanched potatoes shall be immediately cooled with cooling water to the initial temperature of the cooling water and thoroughly drained. The cooled, drained potato dices shall be handled in a manner to prevent discoloration prior to preparation, and filled into the tray pack can within 4 hours after blanching.

3.3.1.2 Preparation of dehydrofrozen potatoes. Dehydrofrozen potatoes shall be fully hydrated until the weight increases by two times, and in such a manner so as to comply with the finished product requirements. The hydrated potatoes shall be handled in a manner to prevent discoloration and shall be filled within 4 hours after hydration.

3.3.2 Preparation of whole eggs. Frozen whole eggs shall be tempered/thawed and held at an internal temperature of 28° to 40°F for not more than 24 hours prior to product preparation.

3.3.3 Preparation of the egg and potato mixture. The egg and potato mixture shall be formulated with the following ingredients:

Ingredient	Percent by weight
Whole eggs, liquid or frozen	51.000
Water	21.154
Potatoes, rehydrated or fresh	11.430
Oil, vegetable	10.710
Starch, waxy maize, modified, pregelatinized, instant	5.200
Salt 1/	0.430
Citric acid	0.070
Color, annatto, dry	0.006

1/ The total amount of salt in the egg/potato mixture shall be adjusted, as necessary, to produce a product that complies with the finished product salt requirements (see 3.6).

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The egg and potato mixture shall be prepared as follows:

a. The vegetable oil and approximately 50 percent of the whole egg amount shall be combined.

b. The starch shall be slowly added to the oil/egg mixture with constant agitation until a smooth blend is formed.

c. The balance of the whole egg amount and the citric acid, salt, and annatto coloring shall then be added to the mixture and mixed until smooth.

d. The water and potatoes shall be uniformly mixed into the egg mixture.

e. The egg/potato mixture shall be included in the omelet preparation (see 3.3.4) immediately after mixing.

3.3.4 Preparation of the omelet. The omelet shall be formulated and prepared as follows:

Ingredient	Percent by weight
Egg/potato mixture	70.0
Sausage links	30.0

a. The sausage links shall be combined with the egg/potato mixture using minimum mixing to form a uniform product.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product to conform to the finished product requirements and to the following requirements:

a. Each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to assure compliance with the finished product requirement (see 3.6p).

b. The filled and sealed tray pack cans shall be in the retort process within 2 hours after preparation.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

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b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. The average net weight shall be not less than 102 ounces.

e. No individual can shall contain less than 100 ounces of product.

f. Free liquid in any individual tray pack can shall be not more than 3.0 ounces.

g. The salt content of the finished product shall be not less than 1.0 percent nor greater than 1.5 percent.

h. The average fat content of the finished product shall not be greater than 21.5 percent and no individual tray pack shall have a fat content greater than 23.0 percent.

i. Potato dice texture shall be firm, but not mushy, hard, fibrous, or tough.

j. At least 75 percent of the potatoes shall be discernible dices.

k. Texture of the omelet shall not be rubbery.

l. Sausage links and potato dices shall be uniformly dispersed throughout the product.

m. The product shall have a typical yellow, cooked egg color.

n. The product shall show no visible lumps of starch.

o. The product shall show no evidence of excessive heating (materially darkened or scorched).

p. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

q. The product shall contain no more than one air pocket or void area measuring 1/2 inch or more in each of two separate dimensions.

r. The product shall contain no more than 2 air pockets or void areas measuring 1/4 inch or more in each of two separate dimensions.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

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3.7 Plant qualification. The meat component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined

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organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to any identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation, or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 102 ounces or if the net weight of any individual can is less than 100 ounces, the lot shall be rejected.

4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated for 40 to 45 minutes in boiling water, opened, and inspected for defects listed in table II.

TABLE I. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3,200	8	-	0	2
	8	16	1	2
3,201 to 35,000	13	-	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

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b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/

Category	Defect
Major	
101	Product shows evidence of excessive heating (materially darkened or scorched).
102	Less than 75 percent of the potatoes are discernible dices.
103	Texture of omelet is rubbery.
104	Sausage links and potato dices not uniformly dispersed throughout the product.
105	Product does not have a typical yellow, cooked egg color.
106	Free liquid weight in any individual tray pack can is more than 3.0 ounces. 3/
107	Texture of potato dice is mushy, hard, fibrous, or tough, not firm. 4/
108	Product shows visible lumps of starch. 5/
109	Presence of 2 or more air pockets or void areas measuring 1/2 inch or more in each of two separate dimensions. 5/
110	Presence of 3 or more air pockets or void areas measuring 1/4 inch or more in each of two separate dimensions. 5/

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1/ The presence of foreign material (e.g. dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ To determine the weight of free liquid the following procedure shall be used. The can lid shall be removed and a portion (approximately 5 inches by 2-3/8 inches) shall be cut out of one corner of the omelet and removed. The lid shall be replaced, leaving approximately 1/2 inch open at the cut out corner. The can shall be elevated on end, so that all liquid will flow out of the cut out corner, and drained for 1 minute into a tared container. The free liquid shall be reported to the nearest 0.1 ounce.

4/ Examination for appearance and texture conformance of the potato dice shall be made following the determination of free liquid.

5/ From each sample can of product, remove one 3 inch wide center slice (sliced lengthwise of the can). Place center slice on edge (including the cut out corner portion) and cut in half lengthwise. Inspect right inside surface for air pockets, void areas and gel lumps.

4.5.3.3 Fat and salt content testing. Nine tray packs shall be selected at random from the lot and distributed as follows:

- Three for laboratory analyses.
- Three for submission to the contractor.
- Three for retention by an AMS agent as reserve samples.

The three tray packs for laboratory analyses shall be individually tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, chapter: Meat and Meat Products, except that preparation of the samples shall be as follows: The three unopened tray packs shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The test results shall be reported to the nearest 0.1 percent. Any test failure is a major defect. The lot shall be rejected if any of the following defects occur:

a. The average fat content of the three tray packs is greater than 21.5 percent.

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b. The fat content of any individual tray pack is greater than 23.0 percent.
c. The salt content of any individual tray pack is less than 1.0 percent or greater than 1.5 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the fat or salt content requirement. When the reserve samples are analyzed, the analyses for both fat and salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United Standards for Condition of Food Containers, except that the inspection for labeling shall be in accordance with 4.5.4.1. In addition, the following defect shall be classified as a critical defect and shall be cause for rejection of the lot:

Evidence of buffing causing damage (i.e. scratches or scuffing) to exterior can coating.

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4) except that, for self-adhering labels, the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can, i.e., label raised or peeled back from edges or corners.

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closure shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and CFR 21, Part 113, Subpart D, or CFR 9, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns is a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to $75^{o} \pm 5^{o}F$, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center

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of the tray pack can. When examining a ribbed lid, only lay the straight edge between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance is a major defect and shall be cause for rejection of the lot.

4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible;
Reinforced with other than nonmetallic strapping or tape;
Dimensions of pads not as specified;
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect, or illegible;
Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

5.1.1 Level A. One hundred and two ounces of food product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5. Buffing of the can causing damage to the exterior can coating is not permitted (see 4.5.4).

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with a HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with

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nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1) except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (e.g., est 38) or a three letter code identifying the establishment.
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

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1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1987 would be coded as 7078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code. Following the four digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined as specified in 4.5.4 after preparation of product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT IN WATER: Submerge unopened can in boiling water. Simmer gently 40 - 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

TO HEAT IN OVEN: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven 35 - 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 2/3 cup each.

As an alternate labeling method, a preprinted self-adhering 0.002 inch thick clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering label is used, the tray pack can shall be labled with the Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE, UNLESS PALLET RACKS ARE USED.

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6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of packing required (see 5.2).
e. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Starch, waxy maize, modified, pregelatinized, instant. It has been found that National 78-0018 Food Starch manufactured by National Starch and Chemical Corporation, Food Products Division, P.O. Box 6500, Bridgewater, NJ, 08807 meets the requirements of 3.2.9 and performs satisfactorily in this product.

6.5 Sausage links, smoked, cooked. It has been found that Oscar Mayer's "Little Smokies" manufactured by Oscar Mayer, P.O. Box 7188, Madison, WI, 53707, Armour's "Cocktail Smokies" manufactured by Armour Food Company, 1230 East Diehl Road, Naperville, IL 60540, and Kahn's "Lit'l Smokies" manufactured by Kahn and Company, 3241 Spring Grove Avenue, Cincinnati, OH 45225 meet the requirements of 3.2.1 and performed satisfactorily during development of this product.

6.6 Subject term (key word) listing.

Canned food
Food processing
Omelet
Potatoes
Ration
Sausage
Thermostabilized
Tray pack

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC
DP - SS

Preparing activity:

Army - GL

Project No. 8940-0654

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-44238A	Omelet with Smoked Sausage and Potatoes, Artificially Colored, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP-1, Feb 86

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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